3 Group Actions

Definition 3.1 (Group Action)**.**
Let $(G, \star)$ be a group and A a non-empty set. A *group action* of G on A is a map:

$$(\cdot) : G \times A \to A$$

$$(g, a) \mapsto g \cdot a$$

satisfying:

(A1) $(g_1 \star g_2) \cdot a = g_1 \cdot (g_2 \cdot a), \quad \forall g_1, g_2 \in G, \quad \forall a \in A.$ **(A2)** $e_G \cdot a = a, \quad \forall a \in A.$

Example 3.1.

1. $(\mathbf{Z}, +)$ acts on $\mathbf{R}$ by translations via the action:

   $$(\cdot) : \mathbf{Z} \times \mathbf{R} \to \mathbf{R}$$

   defined by

   $$(n, r) \mapsto n \cdot r = n + r.$$

   Indeed the axiom (A1) is satisfied:

   $$(n_1 + n_2) \cdot r = (n_1 + n_2) + r = n_1 + (n_2 + r) = n_1 \cdot (n_2 \cdot r)$$

   as well as the axiom (A2):

   $$0 + r = r.$$

2. **Action by left multiplication.** A group $(G, \star)$ acts on itself via the action

   $$(\cdot) : G \times G \to G$$

   defined by

   $$(h, g) \mapsto h \cdot g = h \star g.$$

   Indeed the axiom (A1) is satisfied:

   $$(h_1 \star h_2) \cdot g = (h_1 \star h_2) \star g = h_1 \star (h_2 \star g) = h_1 \cdot (h_2 \cdot g)$$

   as well as the axiom (A2):

   $$e_G \cdot g = e_G \star g = g.$$

   This action is called the *left regular action* of G on itself by left multiplication.

3. **Action by conjugation** A group $(G, \star)$ acts on itself via the action

   $$(\cdot) : G \times G \to G$$

   defined by

   $$(h, g) \mapsto h \cdot g = h \star g \star h^{-1}.$$

   Indeed the axiom (A1) is satisfied:

   $$(h_1 \star h_2) \cdot g = (h_1 \star h_2) \star g \star (h_1 \star h_2)^{-1} = (h_1 \star h_2) \star g \star (h_2^{-1} \star h_1^{-1}) = h_1 \star (h_2 \star g \star h_2^{-1}) \star h_1^{-1} = h_1 \cdot (h_2 \cdot g)$$

   as well as the axiom (A2):

   $$e_G \cdot g = e_G \star g \star e_G^{-1} = g.$$

   This action is called the action of G on itself by conjugation.

Definition (Permutation representation).
Let $(\cdot) : G \times A \to A$ be a group action of a group $(G, \star)$ on the set A. For a fixed $g \in G$ the map

$$\tau_g : A \to A$$

defined by

$$\tau_g(a) = g \cdot a$$

is a bijection. Indeed τ_g has an inverse map $\tau_{g^{-1}} : A \to A$ since

$$\tau_{g^{-1}} \circ \tau_g = \tau_g \circ \tau_{g^{-1}} = \mathrm{id}_A$$

as follows from (A1) and (A2). More precisely:

$$(\tau_{g^{-1}} \circ \tau_g)(a) = \tau_{g^{-1}}(\tau_g(a)) = \tau_{g^{-1}}(g \cdot a) = g^{-1} \cdot (g \cdot a) = (g^{-1} \star g) \cdot a = e_G \cdot a = a$$

for all $a \in A$.

Let $(S_A, \circ)$ be the group of bijections from A to itself; S_A is also called the group of symmetries of A, the group law $\circ$ is the composition of bijections. [Check that $(S_A, \circ)$ is indeed a group.] The map

$$\tau : G \to S_A$$

defined by

$$\tau(g) = \tau_g$$

is a group homomorphism since for $g_1, g_2 \in G$ one has $\tau(g_1 \star g_2) = \tau(g_1) \circ \tau(g_2)$:

$$\tau(g_1 \star g_2)(a) = \tau_{g_1 \star g_2}(a) = (g_1 \star g_2) \cdot a = g_1 \cdot (g_2 \cdot a) = \tau_{g_1}(\tau_{g_2}(a)) = (\tau(g_1) \circ \tau(g_2))(a)$$

for all $a \in A$. The homomorphism τ is called the *permutation representation* associated to the given action $(\cdot)$.

Definition 3.2 (Kernel of a Representation).
The kernel of $\tau : G \to S_A$

$$\mathrm{Ker}(\tau) = \{g \in G : \tau(g) = \tau_g = \mathrm{id}_A\} = \{g \in G : g \cdot a = a, \forall a \in A\},$$

is called the *kernel of the representation* τ, it is a normal subgroup of G. If $\mathrm{Ker}(\tau) = \{e_G\}$, or equivalently if τ is injective, we say that the action $(\cdot)$ is *faithful*.

Example 3.2.

1. The left regular representation of G on itself arising from the action of G on itself by left multiplication (see Examples 3.1) is faithful. Indeed let $g \in G$ with $\tau_g = \mathrm{id}_G$, meaning $\tau_g(h) = g \cdot h = h, \forall h \in G$. For $h = e_G$ we obtain $g = g \cdot e_G = e_G$.
2. Consider the action of D_8 on itself by conjugation and the corresponding permutation representation $\tau : D_8 \to S_{D_8}$. This action is not faithful. Indeed $r^2 \in \mathrm{Ker}(\tau)$ since

   $$r^2 s^i r^j (r^2)^{-1} = r^2 s^i r^j r^2 = s^i r^2 r^j r^2 = s^i r^{4+j} = s^i r^j$$

   holds for each $s^i r^j \in D_8$, hence $\tau(r^2) = \mathrm{id}_{D_8}$.

3. Assume G is abelian and consider the action of G on itself by conjugation and the corresponding representation $\tau : G \to S_G$. Then this action is not faithful if $|G| > 1>$. In fact $\mathrm{Ker}(\tau) = G$ as $\tau(g) = \tau_g = \mathrm{id}_G, \forall g \in G$:

$$\tau(g)(h) = \tau_g(h) = g \star h \star g^{-1} = g \star g^{-1} \star h = h$$

for all $h \in G$.

Definition 3.3 (Stabiliser)**.**
Let $G \times A \to A$ be an action of a group $(G, \star)$ on a set A and $a \in A$. The *stabiliser* of a is

$$\mathrm{Stab}(a) = \{g \in G : g \cdot a = a\}.$$

Proposition 3.1.
The stabiliser $\mathrm{Stab}(a)$ *is a subgroup of* G*.*

Proof. First $e_G \in \mathrm{Stab}(a)$ since $e_G \cdot a = a$. Let $g \in \mathrm{Stab}(a)$ meaning $g \cdot a = a$. Then

$$a = e_G \cdot a = (g^{-1} \star g) \cdot a = g^{-1} \cdot (g \cdot a) = g^{-1} \cdot a$$

and $g^{-1} \in \mathrm{Stab}(a)$. Further, let $g_1, g_2 \in \mathrm{Stab}(a)$ meaning $g_1 \cdot a = g_2 \cdot a = a$ then

$$(g_1 \star g_2) \cdot a = g_1 \cdot (g_2 \cdot a) = g_1 \cdot a = a$$

and $g_1 \star g_2 \in \mathrm{Stab}(a)$. □

Example 3.3.

1. In the example 1 of Examples 3.1 let $r \in \mathbf{R}$, then

$$\mathrm{Stab}(r) = \{n \in \mathbf{Z} : n + r = r\} = \{0\}.$$

2. Consider the action of S_3 on the set $\{1, 2, 3\}$ by permutations. Then $\mathrm{Stab}(1) = \{e, (23)\}$, $\mathrm{Stab}(2) = \{e, (13)\}$, and $\mathrm{Stab}(3) = \{e, (12)\}$.

Let $(G, \star)$ be a group acting on a non-empty set A. Define a relation R among the elements of A by:

$$a \ R \ b \iff \exists g \in G : a = g \cdot b$$

for $a, b \in A$.

Proposition 3.2.
The relation R *is an equivalence relation.*

Proof. R is reflexive since $a = e_G \cdot a$ hence $a \ R \ a$, $\forall a \in A$. Let $a, b \in A$ with $a \ R \ b$, i.e. $a = g \cdot b$ for some $g \in G$. Then $g^{-1} \cdot a = g^{-1} \cdot (g \cdot b) = (g^{-1} \star g) \cdot b = e_G \cdot b = b$ hence $b \ R \ a$. Finally, let $a, b, c \in A$ with $a \ R \ b$ and $b \ R \ c$. Thus $\exists g, g' \in G$ such that $a = g \cdot b$ and $b = g' \cdot c$. Then $a = g \cdot b = g \cdot (g' \cdot c) = (g \star g') \cdot c$ and $a \ R \ c$. □

Definition 3.4 (Orbit).
Let $a \in A$. The equivalence class of a for the relation R is

$$\bar{a} = \{b \in A : b \ R \ a\} = \{b \in A : \exists g \in G, \ b = g \cdot a\} = \{g \cdot a : g \in G\}.$$

It is called the *orbit* of a, for the given action, and is denoted $\mathrm{orb}(a)$.

Note that A is equal to the disjoint union of its orbits under the given action of G:

$$A = \bigcup_{a \in A} \mathrm{orb}(a).$$

The action of G on A is called *transitive* if there is only one orbit, in which case this orbit necessarily contains all elements of A, i.e. $\forall a, b \in A$, $\exists g \in G$ with $a = g \cdot b$. Thus $A = \mathrm{orb}(a)$ for every $a \in A$ in this case.

Example 3.4.

1. The action
$$S_n \times \{1, 2, \ldots, n\} \to \{1, 2, \ldots, n\}$$
$$(\sigma, i) \mapsto \sigma(i)$$
of S_n on the set $\{1, 2, \ldots, n\}$ is transitive: given $1 \leq i \leq j \leq n$ we can find $\tau \in S_n$ with $\tau(i) = j$. For example take $\tau = \mathrm{id}$ if $i = j$, or $\tau = (ij)$ otherwise.
2. The group D_8 acts transitively on the four vertices of a square and the stabiliser of any vertex is the subgroup of order 2 (and index 4) consisting of identity and the reflection about the line of symmetry passing through that point.

Theorem 3.3 (The Orbit-Stabiliser Theorem).
Assume $(G, \star)$ is a finite group acting on a set A. Then the orbit $\mathrm{orb}(a)$ of an element $a \in A$ is finite and

$$|\mathrm{orb}(a)| = |G : \mathrm{Stab}(a)| = \frac{|G|}{|\mathrm{Stab}(a)|}.$$

Proof. Consider the map

$$f : \mathrm{orb}(a) \to (G/\mathrm{Stab}(a))_{\mathrm{left}}$$

defined by

$$f(g \cdot a) = g.\,\mathrm{Stab}(a).$$

We check this map is well defined, i.e.

$$\{g_1 \cdot a = g_2 \cdot a\} \Rightarrow \{g_1.\,\mathrm{Stab}(a) = g_2.\,\mathrm{Stab}(a)\}.$$

We have $(g_2^{-1} \star g_1) \cdot a = g_2^{-1} \cdot (g_1 \cdot a) = g_2^{-1} \cdot (g_2 \cdot a) = (g_2^{-1} \star g_2) \cdot a = e_G \cdot a = a$ hence $g_2^{-1} \star g_1 \in \mathrm{Stab}(a)$ which implies $g_1.\,\mathrm{Stab}(a) = g_2.\,\mathrm{Stab}(a)$.

Further f is a bijection. Indeed f is clearly surjective, we prove it is injective. Suppose $g_1 \cdot a, g_2 \cdot a \in \mathrm{orb}(a)$ and $f(g_1 \cdot a) = f(g_2 \cdot a)$, i.e. $g_1.\,\mathrm{Stab}(a) = g_2.\,\mathrm{Stab}(a)$ which is equivalent to $g_2^{-1} \star g_1 \in \mathrm{Stab}(a)$. Then $a = (g_2^{-1} \star g_1) \cdot a = g_2^{-1} \cdot (g_1 \cdot a) \Rightarrow g_2 \cdot a = g_2 \cdot (g_2^{-1} \cdot (g_1 \cdot a)) = (g_2 \star g_2^{-1}) \cdot (g_1 \cdot a) = e_G \cdot (g_1 \cdot a) = g_1 \cdot a$ as required.

Now since f is a bijection, $\mathrm{orb}(a)$ and $(G/\operatorname{Stab}(a))_{\text{left}}$; whose cardinality is $|G : \operatorname{Stab}(a)|$, have the same number of elements. The statement follows using Lagrange's theorem which asserts that $|G : \operatorname{Stab}(a)| = \frac{|G|}{|\operatorname{Stab}(a)|}$. □

Example 3.5.

- Consider the action of S_3 on itself by conjugation. Then

$$\operatorname{Stab}(12) = \{e, (12)\}$$

and

$$\mathrm{orb}(12) = \{(23) = (123)(12)(132), (13) = (132)(12)(123), (12) = e(12)e^{-1}\}.$$

Further

$$\operatorname{Stab}(123) = \{1, (123), (132)\}$$

and

$$\mathrm{orb}(123) = \{(123) = e(123)e^{-1}, (132) = (12)(123)(12)\}.$$

Cayley's Theorem. Recall the left regular representation of $(G, \star)$ on itself by left multiplication (see example 2 in Examples 3.1). Assume that G is finite of cardinality n. If we label the elements of G as $\{g_1, \ldots, g_n\}$, the regular representation defines a faithful permutation representation (an injective homomorphism)

$$\tau : G \to S_n$$

called the *regular permutation representation*, defined as follows: if $g \in G$, $\tau(g)$ is the permutation defined for $i, j \in \{1, \ldots, n\}$ by

$$\tau(g)(i) = j \quad \text{if} \quad g \star g_i = g_j.$$

The permutation representation τ depends on the given labelling of the elements of G.

In particular, since $\operatorname{Ker}(\tau) = \{e_G\}$ we obtain by the FIT that G is isomorphic to its image $\tau(G)$; a subgroup of S_n. We thus obtain.

Theorem 3.4 (Cayley's Theorem).
A finite group of cardinality n is isomorphic to a subgroup of S_n.

Next, we define the left action of a group $(G, \star)$ on the set of left cosets of a given subgroup. Let H be a subgroup of G, and $A = (G/H)_{\text{left}}$ the set of left cosets of H. The group G acts on A by

$$g \cdot (g'H) = (g \star g')H$$

(check that the group action axioms are satisfied). This action is called the action of G on the left cosets of H by left multiplication.

If $|G : H| = m$ is finite, and we label the elements of A as $\{g_1H, \ldots, g_mH\}$, then the above representation defines a homomorphism

$$\tau : G \to S_m$$

as follows: if $g \in G$, $\tau(g)$ is the permutation defined for $i, j \in \{1, \ldots, m\}$ by

$$\tau(g)(i) = j \quad \text{if} \quad g \cdot g_iH = (g \star g_i)H = g_jH.$$

The permutation representation τ depends on the given labelling of the elements of A.

Example 3.6.
Let $G = D_8$ and $H = \langle s \rangle = \{1, s\}$. Label the distinct left cosets H, rH, r^2H, and r^3H of H with the integers 1, 2, 3, and 4 respectively. Under this labelling we compute the permutation σ_s induced by the action by multiplication by the element s on the left cosets of H:

$$s \cdot H = \{s, s^2\} = H$$

so $\sigma_s(1) = 1$,

$$s \cdot (rH) = srH = \{sr, srs\} = \{r^3s, r^3\} = r^3H$$

so $\sigma_s(2) = 4$,

$$s \cdot r^2H = sr^2H = \{sr^2, sr^2s\} = \{r^2s, r^2s^2 = r^2\} = \{r^2s, r^2\} = r^2H$$

so $\sigma_s(3) = 3$, and

$$s \cdot r^3H = sr^3H = \{sr^3, sr^3s\} = \{rs, rs^2\} = \{rs, r\} = rH$$

so $\sigma_s(4) = 2$. Hence for this labelling $\sigma_s = (24)$. Similarly one can compute, under this labelling, the permutation σ_r induced by the action by multiplication by the element r on the left cosets of H to find $\sigma_r = (1234)$.

Let

$$\tau_H : G \to S_{G/H}$$
$$g \mapsto \tau_H(g)$$

be the permutation representation associated to the action of G by left multiplication on the left cosets of H.

Thus if $g \in G$,

$$\tau_H(g) : G/H \to G/H$$

is the bijection defined by

$$\tau_H(g)(g'H) = (g \star g')H.$$

Theorem 3.5.
With the notations above, the followings hold.

1. *G acts transitively on G/H.*
2. *The stabiliser of e_GH is the subgroup H.*
3. *$\mathrm{Ker}(\tau_H) = \bigcap_{x \in G} xHx^{-1}$, and $\mathrm{Ker}(\tau_H)$ is the largest normal subgroup of G contained in H.*

Proof.

1. To see that G acts transitively on G/H let $aH, bH \in G/H$ and $g = b \star a^{-1}$. Then $g \cdot (aH) = (b \star a^{-1}) \cdot aH = (b \star a^{-1} \star a)H = bH$ this proves 1.
2. The stabiliser of e_GH is $\{g \in G : g \cdot (e_GH) = gH = H\} = \{g \in G : gH = H\} = H$, this shows 2.
3. Next, we prove 3. By definition $\mathrm{Ker}(\pi_H) = \{g \in G : g \cdot (xH) = xH, \forall x \in G\} = \{g \in G : (g \star x)H = xH, \forall x \in G\} = \{g \in G : (x^{-1} \star g \star x)H = H, \forall x \in G\} = \{g \in G : x^{-1} \star g \star x \in H, \forall x \in G\} = \{g \in G : g \in xHx^{-1}, \forall x \in G\} = \bigcap_{x \in G} xHx^{-1}$. Further $\mathrm{Ker}(\pi_H)$ is contained in $H = e_GHe_G^{-1}$, and $\mathrm{Ker}(\pi_H)$ is a normal subgroup of both G and H. Now let N be a normal subgroup of G contained in H then $N = xNx^{-1} \subset xHx^{-1}$, $\forall x \in G$, hence $N \subset \bigcap_{x \in G} xHx^{-1} = \mathrm{Ker}(\pi_H)$. This shows $\mathrm{Ker}(\pi_H)$ is the largest normal subgroup of G contained in H.

□

Corollary 3.6.
Let G be a finite group of cardinality $n \geq 2$ and p the smallest prime number dividing $n = |G|$. Then any subgroup of G of index p is normal. In particular, if G has a subgroup of index 2 then this subgroup must be normal.

Proof. Suppose $H \leq G$ is a subgroup (the notation $\leq$ is used in the literature to denote a subgroup) and $|G : H| = p$. Let π_H be the permutation representation afforded by multiplication on the set of left cosets of H in G. Let $K = \mathrm{Ker}(\pi_H)$; K is contained in H, and $|H : K| = m$. Then $|G : K| = |G : H||H : K| = pm$ (use Lagrange's theorem for the subgroup H of G and the subgroup K of H to show this).

Since H has p left cosets in G, G/K is isomorphic to a subgroup of S_p (namely the image of G under π_H) by the FIT. By Lagrange's Theorem, $pm = |G/K|$ divides $|S_p| = p!$. Thus m divides $\frac{p!}{p} = (p-1)!$. But all prime factors of $(p-1)!$ are less than p, and by the minimality of p, every possible prime divisor of m is greater or equal to p. This forces $m = 1$ as m divide $|G|$. So $H = K$ is a normal subgroup of G (since K is normal): the equality $|H : K| = 1$ just means that $H = K$. □